2026 Tanner Crab CIE Review
Spatial Considerations

William Stockhausen

AFSC/NMFS/NOAA

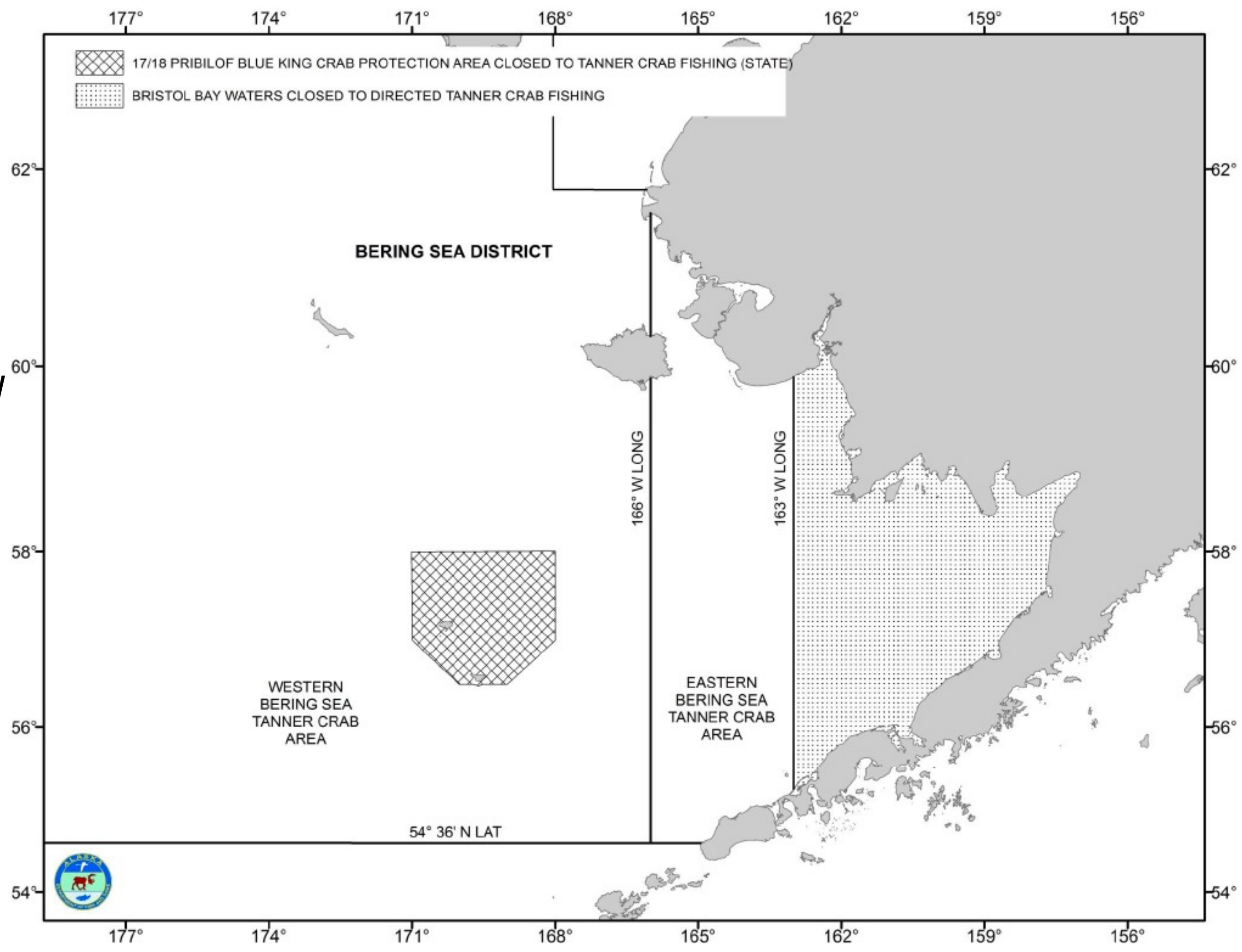
Stock structure and spatial considerations

- For federal management purposes,
  Tanner crab is considered one stock in the eastern Bering Sea
- Somerton (1980) and subsequent work hypothesized two stocks
  - 166^{o}W longitude as dividing line
  - Based on area-specific differences in aggregated characteristics
    - mean size, weight
    - size-at-maturity
    - spatial patterns of abundance and biomass
  - Prior to recognition of male terminal molt
- Spatial patterns result of
  - stock structure?
  - movement?
- Evidence for stock structure?
  - Johnson (2019) genetics study found no structure
  - no concerted tagging studies
- But…

Tanner crab fishery management

Federal
- Considered one stock
- Single OFL, ABC
- Single stock assessment model

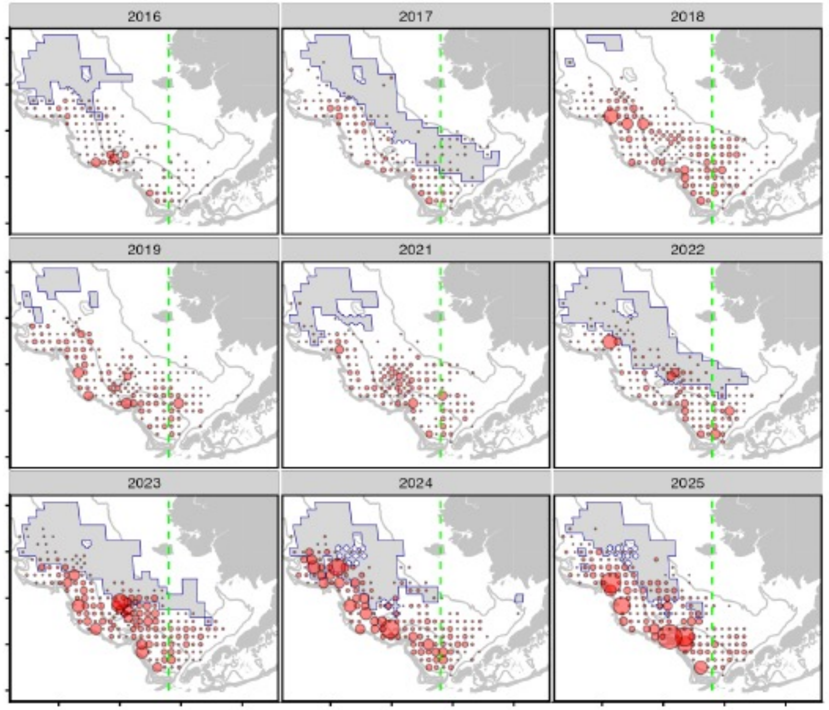
State
- Sets TACs separately for areas east and west of 166°W
  - required: sum(TACs) < ABC
- Area-specific harvest control rules
  - mature female biomass for entire EBS
  - area-specific mature male biomass
    - east: > 113 mm CW
    - west: > 103 mm CW
  - area-specific 5-inch males
    - abundance
    - average weight
    - new shell/old shell proportions



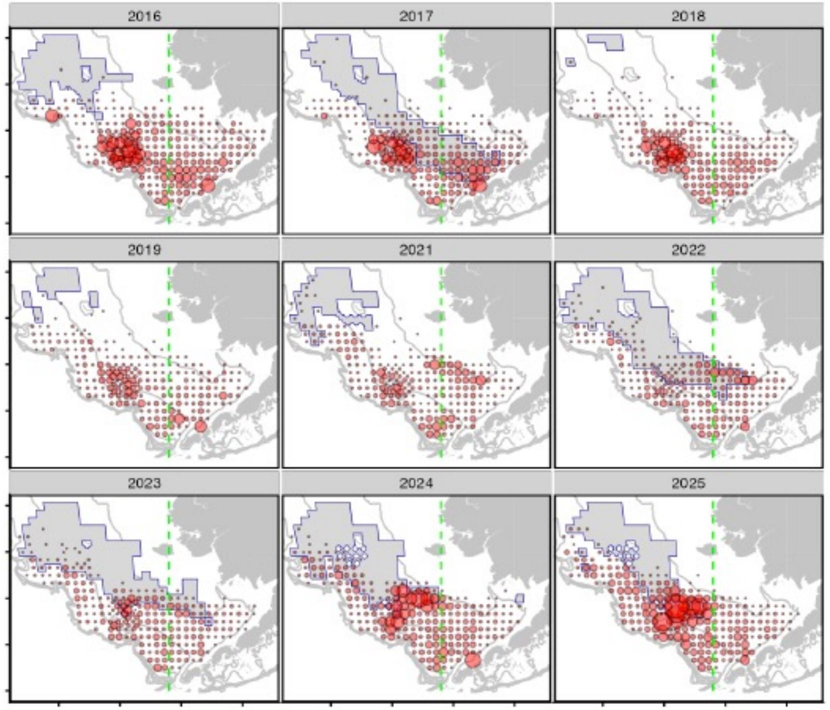
Courtesy of Ben Daly, ADFG

Survey Spatial Patterns

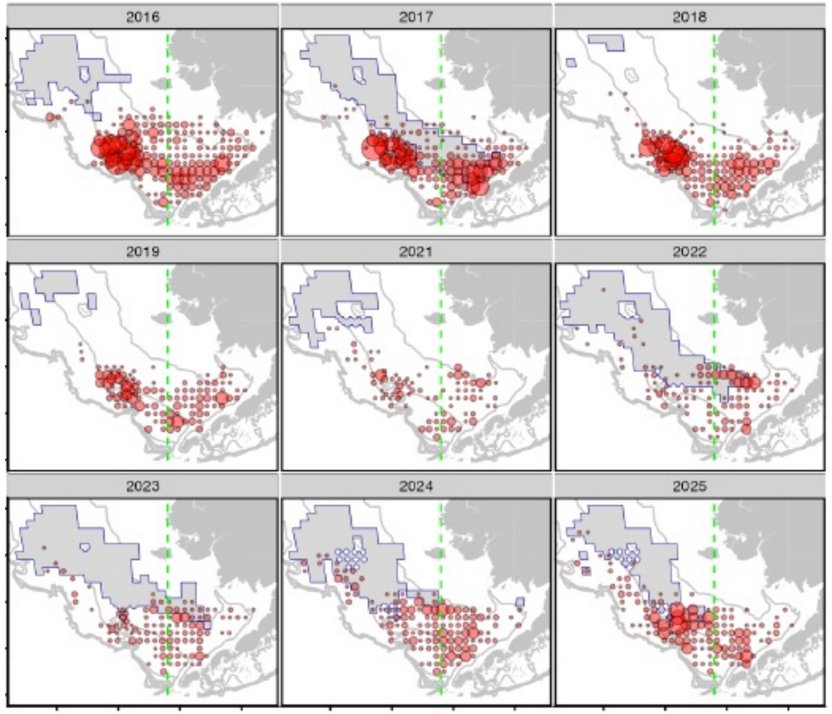
small males (< 60 mm CW)



large males (> 60 mm CW)



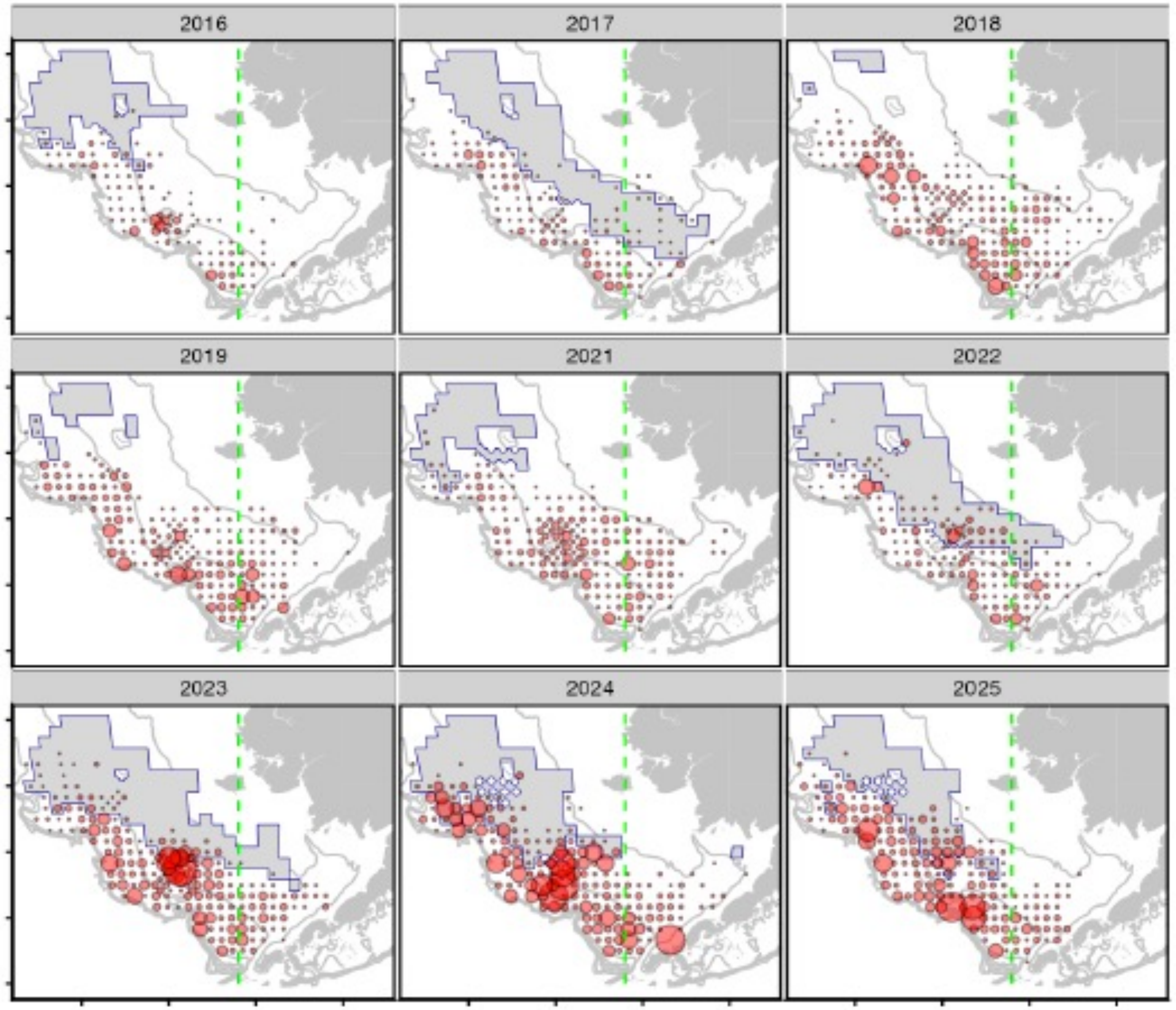
IPMs



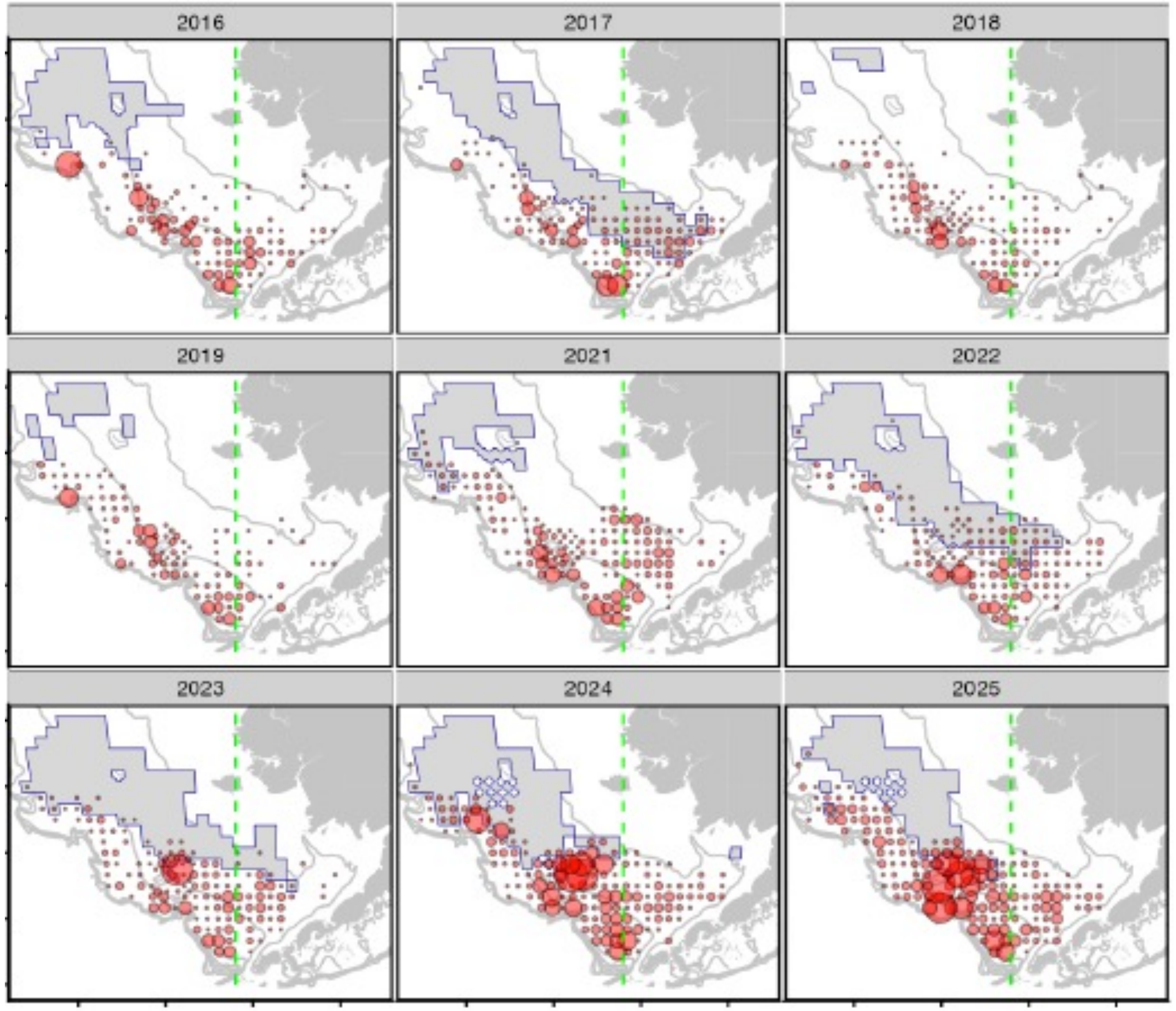
different groups: different scales

Survey Spatial Patterns

immature females



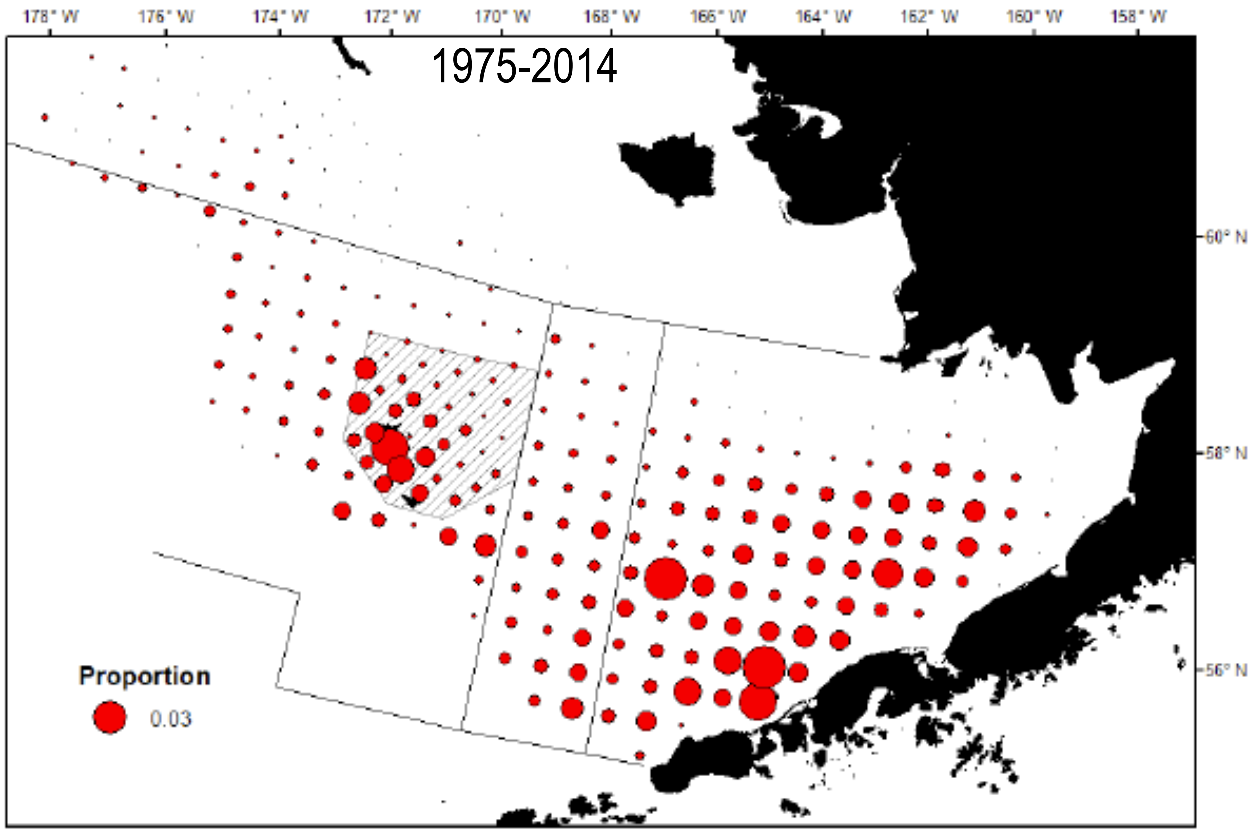
mature females

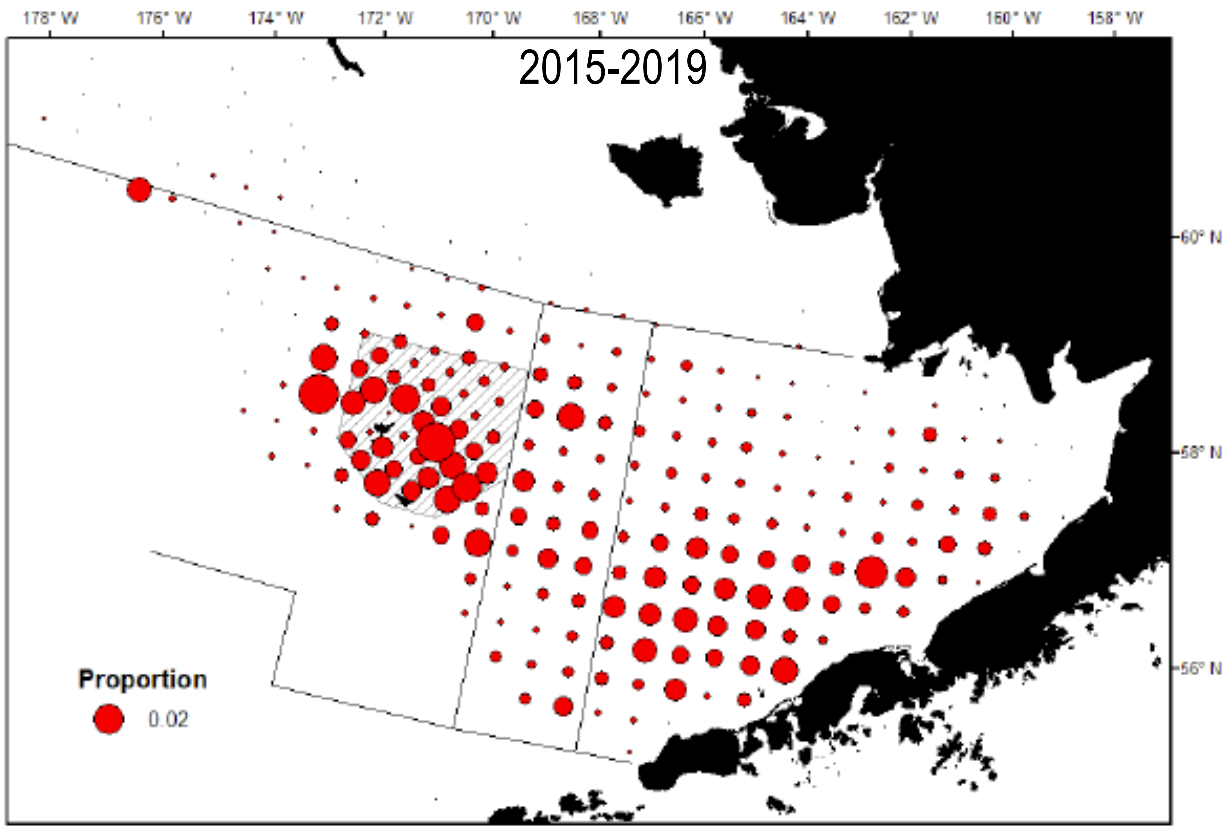


different groups: different scales

NOAA FISHERIES

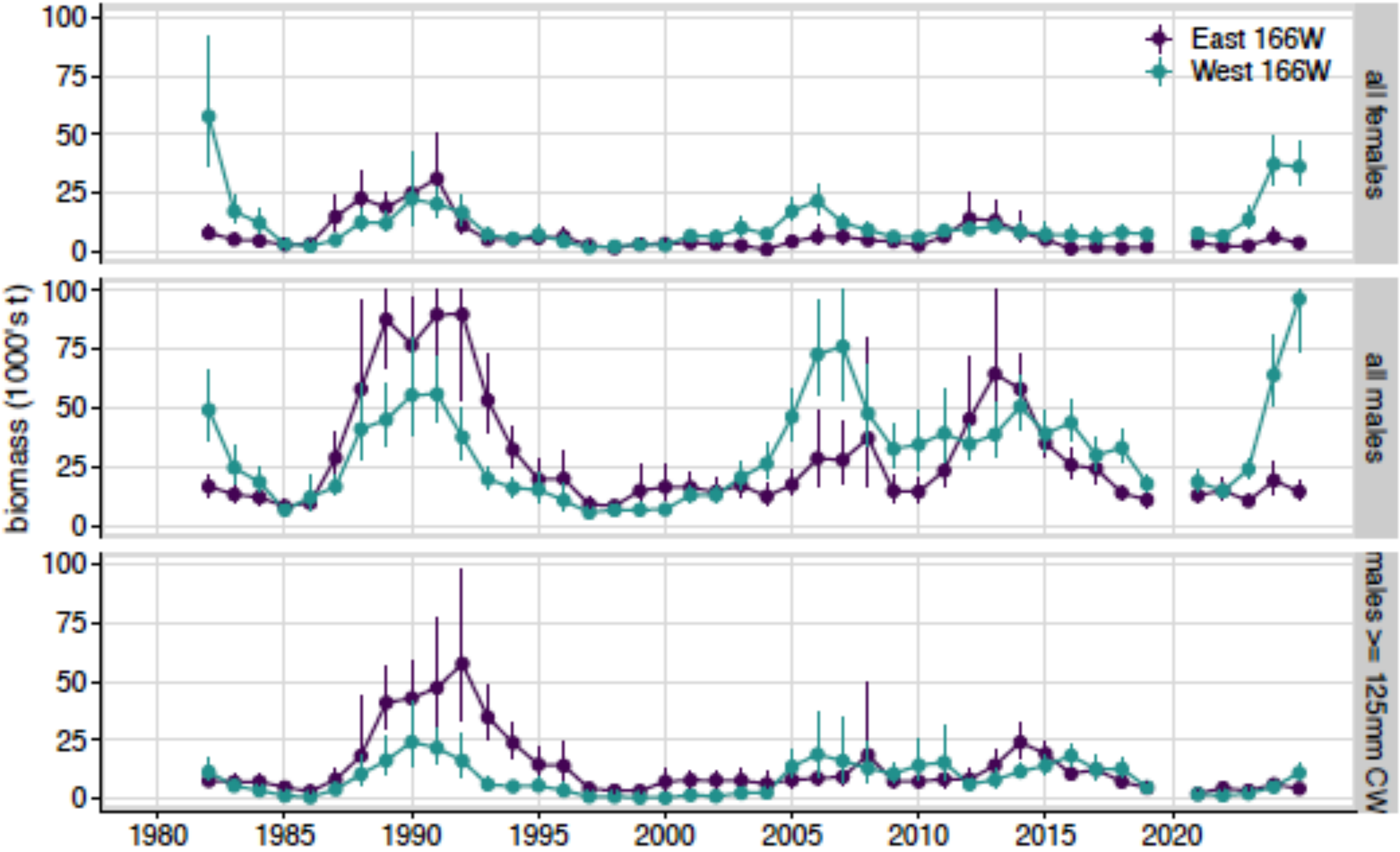
East/West Differences: Mature Male Biomass



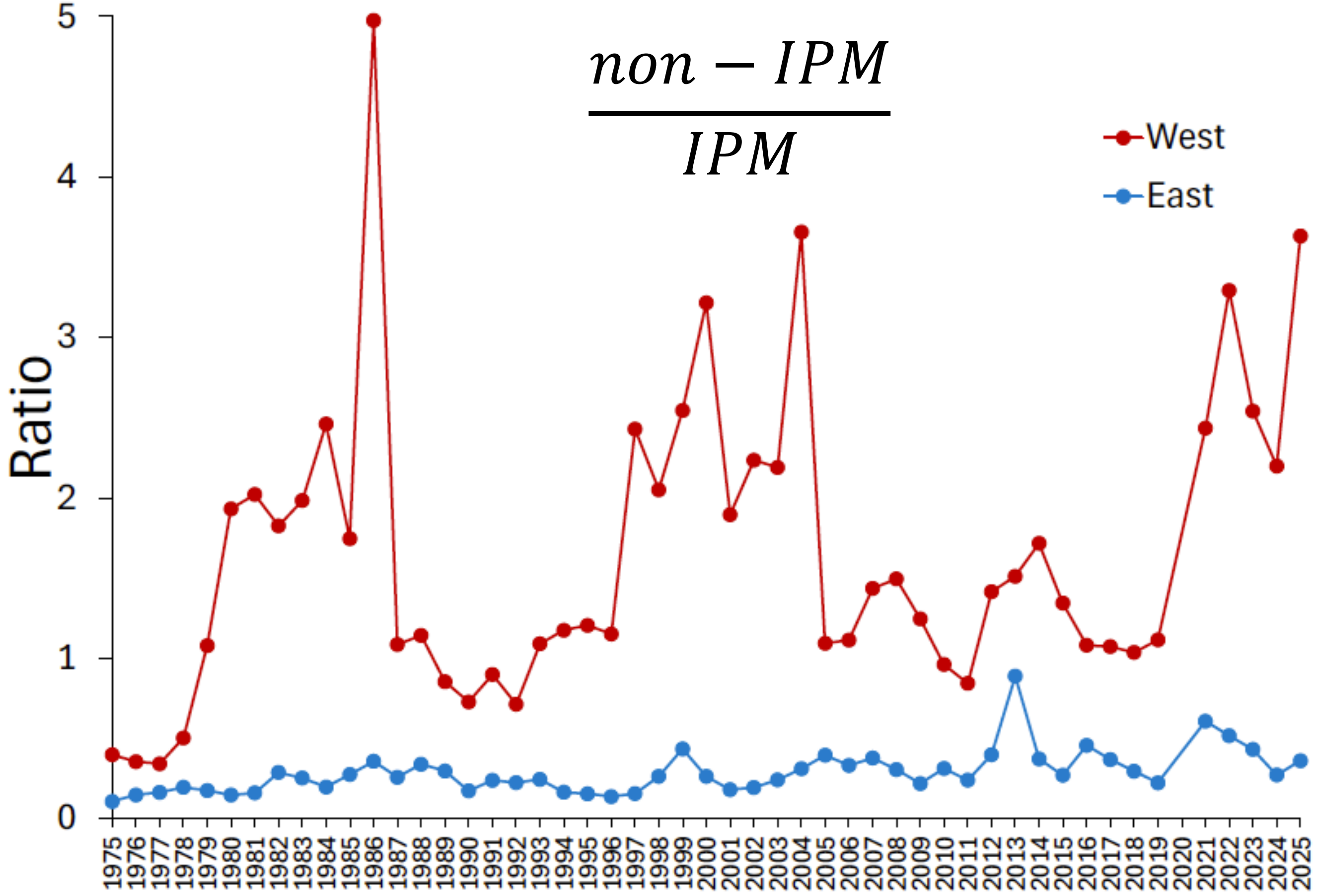


Courtesy of Ben Daly, ADFG

Survey Data By Management Region

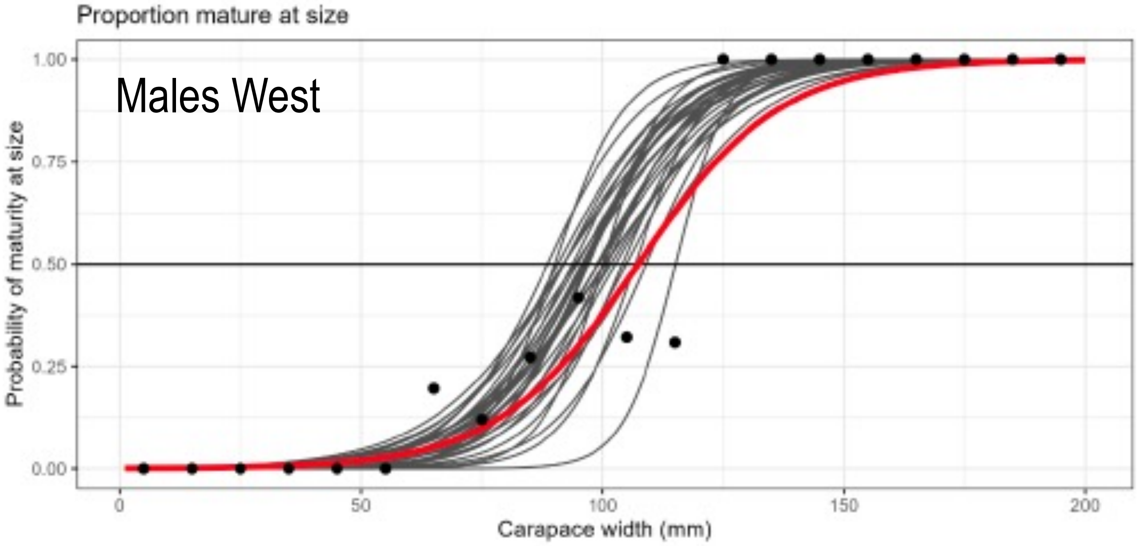


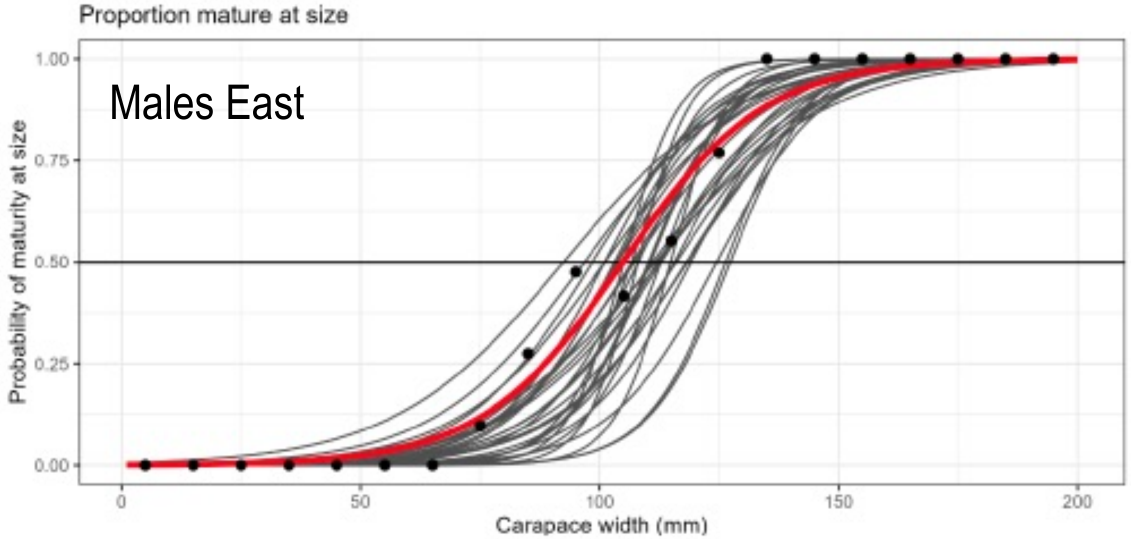
East/West Differences



Courtesy of Ben Daly, ADFG

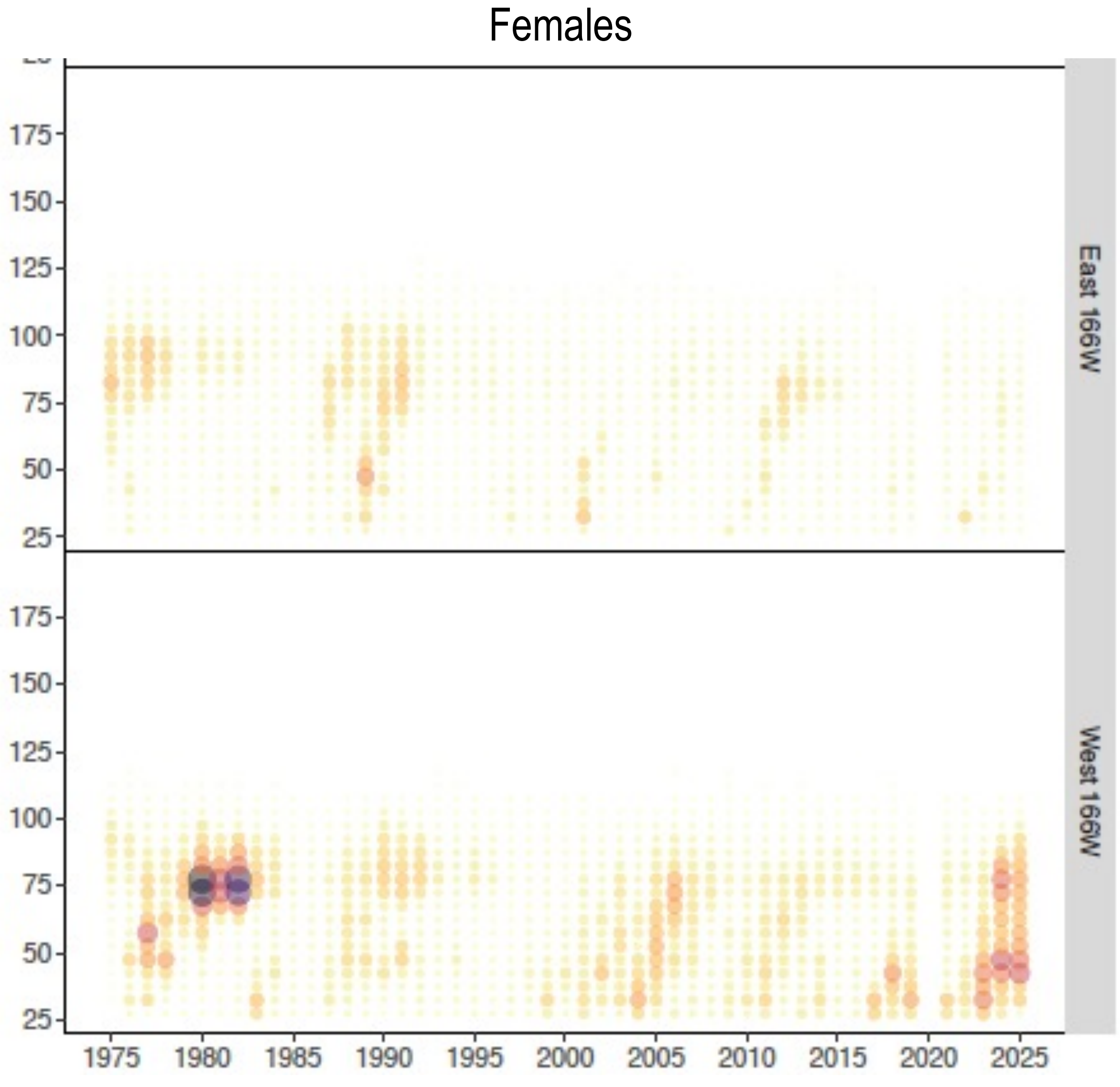
IPM: Industry-preferred males (>125 mm CW)

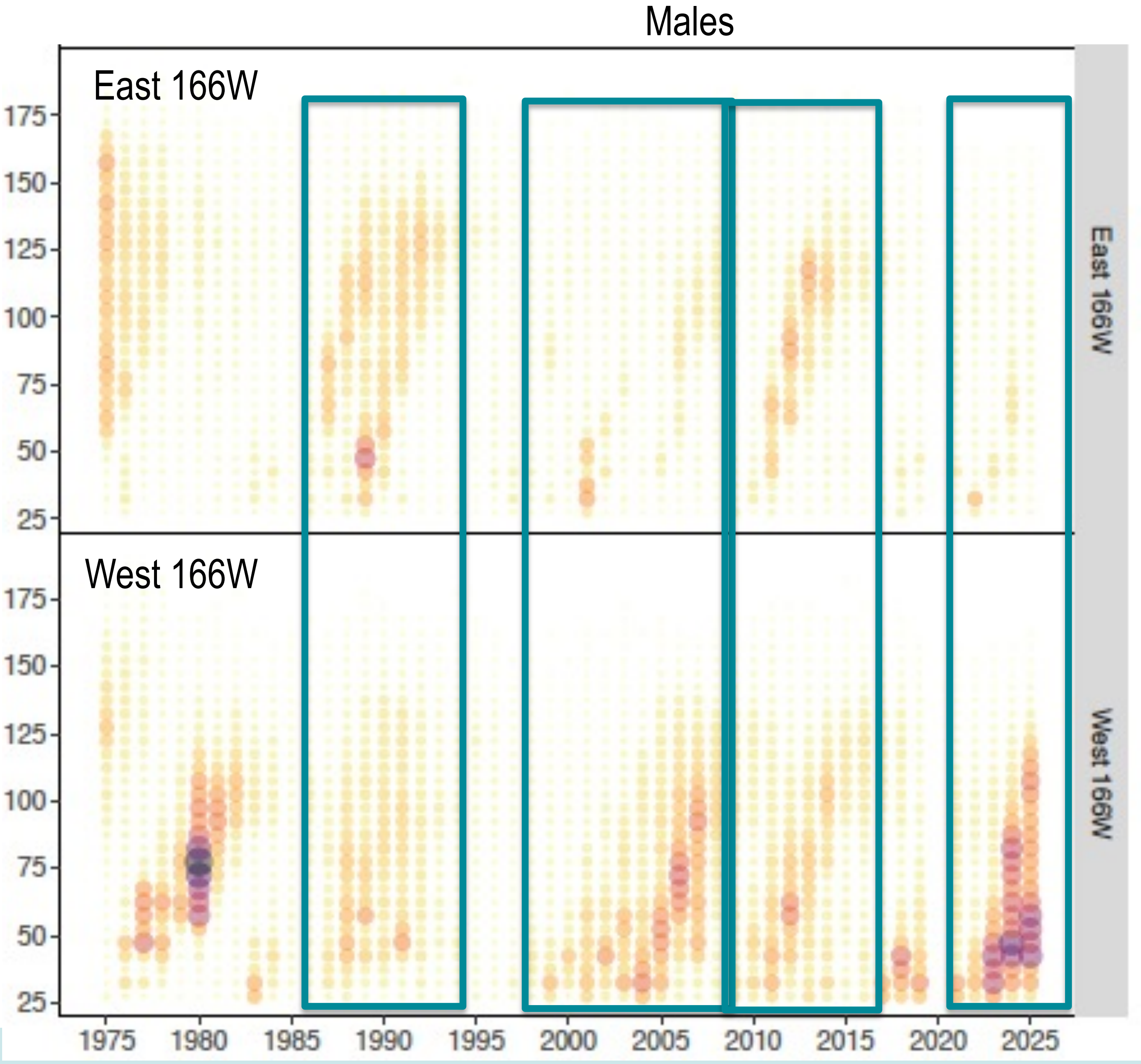




2025 Survey Tech Memo

Survey Size Comps: Synchronous recruitment?

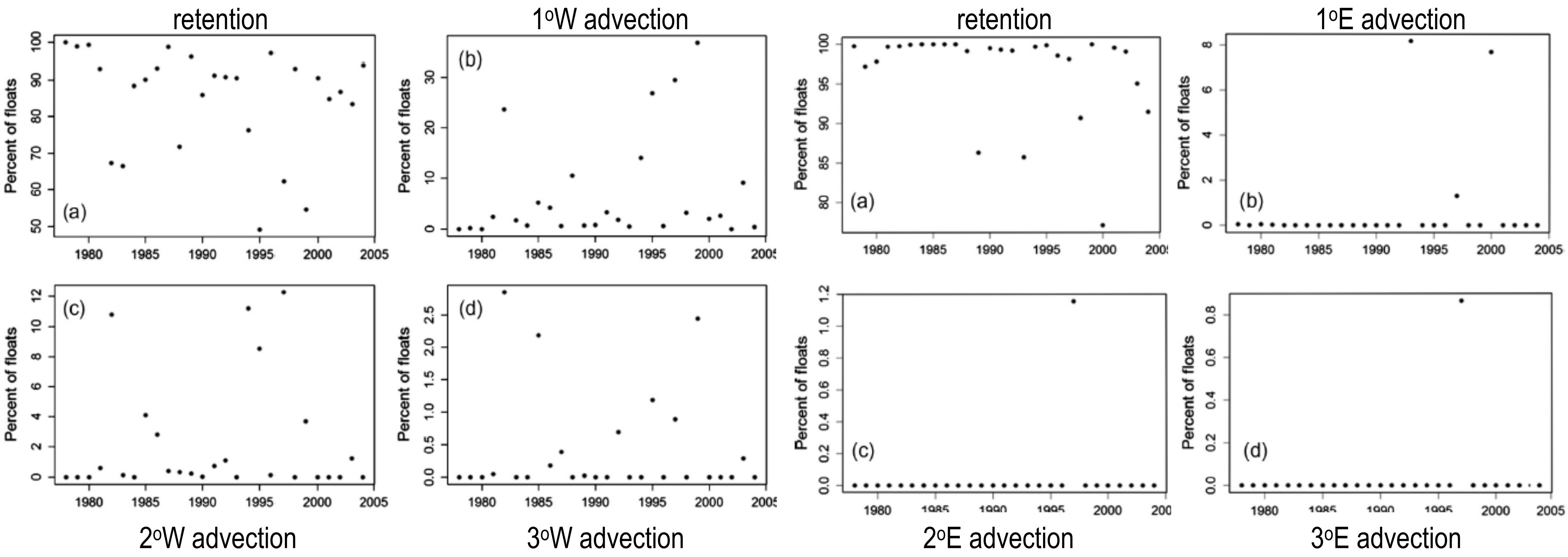




Asymmetric larval transport?

Originating East of 166W

Originating West of 166°W



From Richar et al. 2015

Spatial Patterns and Stock Structure

- Spatial patterns exist: do they suggest multiple stocks in the EBS?
  - Johnson (2019) genetics study suggests no
- Spatial patterns exist: do they suggest the need for a spatial assessment?
- Suggested research directions?
- Priorities?